F78AB - Actuarial and Financial Mathematics B
Computer Lab 3 – Force of interest

Question 1 A payment of £2000 is paid at the end of every year for 10 years.

(a) Show the cashflow occurring in each year in a spreadsheet.

The constant force of interest is 4% p.a.

(b) Calculate the present value of each cashflow at the start of the payment stream.

(c) Hence calculate the total present value of the cashflows. *Ans: £16 156.52.*

Question 2 Payments to contractors on a tram project are made monthly in advance for 15 months, and increase by 3% per month effective. The initial monthly payment is £4 million.

(a) Show the cashflow occurring in each month in a spreadsheet.

The constant force of interest is 1% per month.

(b) Calculate the present value of each cashflow at the start of the payment stream.

(c) Hence calculate the total present value of the cashflows. *Ans: £69.050 million.*

Question 3

Time is measured in years. At time $t = 0$, an investor believes that the annual force of interest $\delta(t)$ on a savings account will be

$$\delta(t) = \begin{cases} 0.03 & \text{for } 0 \leq t \leq 4 \\ 0.05 & \text{for } t > 4. \end{cases}$$

The investor plans to deposit £10 000 in the savings account at the end of every year for the next 20 years.

(a) Show the cashflow occurring in each year in a spreadsheet.

(b) Calculate the discount factor to be applied to each cashflow, so that each cashflow is discounted back to $t = 0$.

(c) Calculate the present value of each cashflow.

(d) Hence calculate the total present value of the cashflows. *Ans: £132 389.24.*